Linear Mixed Models

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Restricted maximum likelihood (REML)

Problem: ML estimates of variance components are biased, because we also have to estimate β

REML solution: Write down a new likelihood function for a transformation of $\mathbf{y}$ that does *not* depend on β

REML

Model assumptions

$$\begin{aligned}\text{Heterozygosity}_{ij} = \beta_0 + u_i + \beta_1 \log(\text{breeding range}_{ij}) \\ + \beta_2 \log(\text{mass}_{ij}) + \beta_3 \text{latitude}_{ij} + \varepsilon_{ij}\end{aligned}$$

$$u_i \overset{iid}{\sim} N(0, \sigma_u^2) \qquad \varepsilon_{ij} \overset{iid}{\sim} N(0, \sigma_\varepsilon^2)$$

Question: What assumptions is this model making?

Types of residuals

Upwards residual analysis